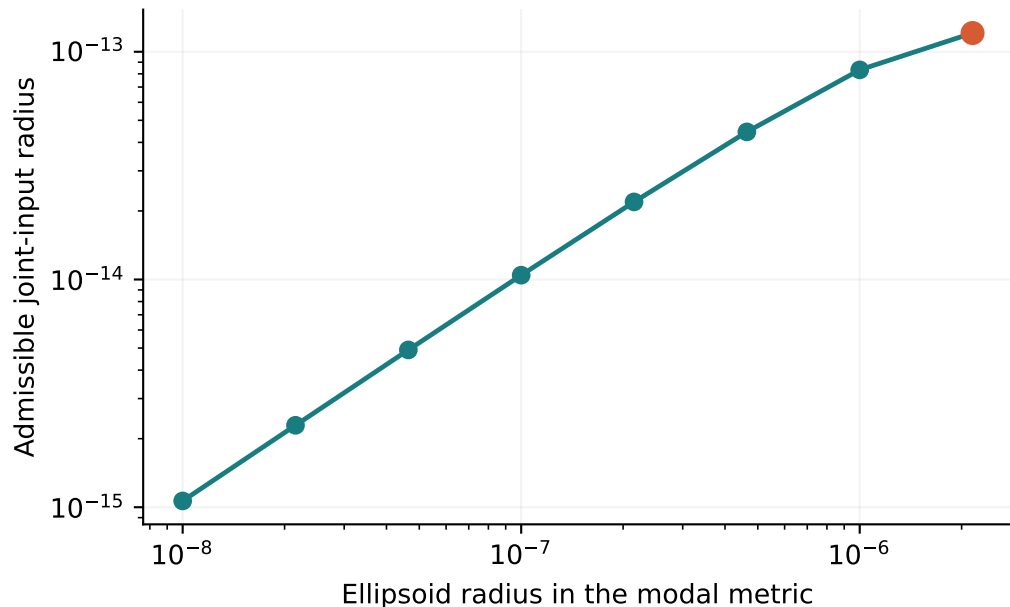


Coupled nonlinear certificate: measuring the loss from conservatism

281 states · all 39 bus instruments · 88 disturbance channels · fixed SG/GFL design

A Positive, but practically negligible envelope



The full nonlinear DAE is enclosed throughout the region. A failed bound does not demonstrate instability.

No practical replacement capacity, globally optimal gains, or hardware safety is established by this microscopic certificate.

B Point checks do not replace a regional bound

